

Conversations in Space: Non-Linear LLM Interaction in Everyday Use

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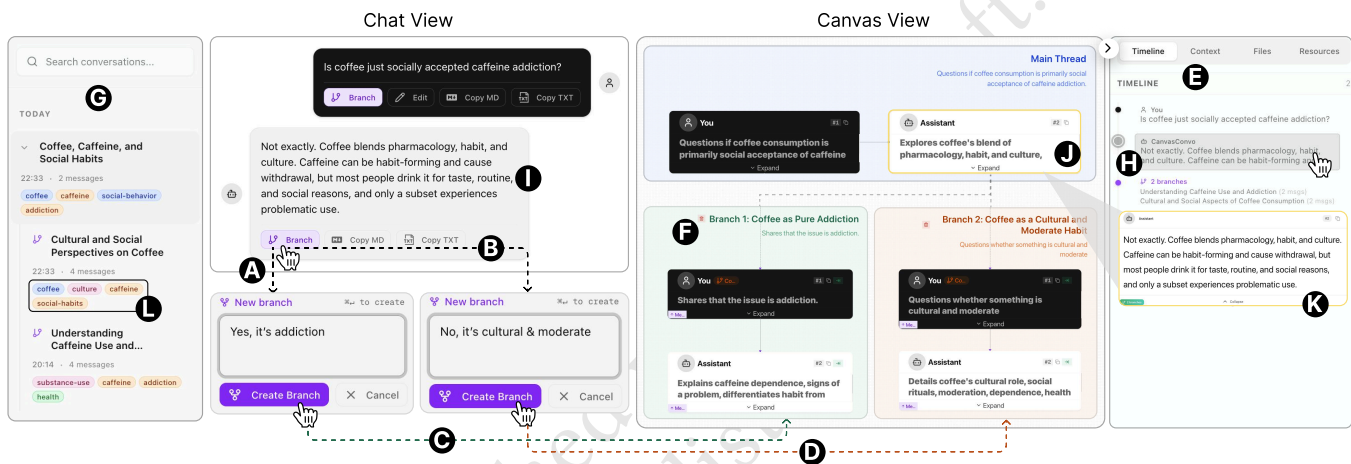


Figure 1: Our system CanvasConvo extends traditional chat-based interaction (Chat View) with branching and spatial organization (Canvas View). Starting from a single prompt, users can create alternative conversational paths directly from messages (A, B), enabling parallel exploration of different perspectives. The canvas view represents these branches as a structured node-link graph, supporting comparison, navigation, and revisiting of ideas (C, D). The timeline (E) is linked to both linear and spatial views: selecting an entry (H) scrolls to the corresponding point in the chat (I), while in the canvas, it highlights the referenced node (J) and enables users to zoom into its details on click (K). The sidebar (G) organizes conversations with tags (L).

Abstract

As LLM conversations grow, their histories capture alternative directions, decisions, and evolving lines of thought that can be difficult to navigate through chat alone. We investigate an interaction concept that represents the same conversation through two synchronized views: a familiar linear chat for ongoing dialogue and a spatial canvas for navigating its emerging structure. To investigate this interaction concept, we developed CanvasConvo, which allows conversations to branch into alternative paths that remain accessible across both views. In a five-day field deployment with 24 participants, we examined how people appropriated this parallel representation in self-directed knowledge work. Participants selectively moved between the two views rather than replacing chat with the canvas. Chat remained central to conversational interaction, while the canvas supported overview, revisitation, and exploration of alternatives. Adoption was uneven, revealing challenges around

established chat habits, transitions between representations, and understanding branch context. Our findings inform the design of user interfaces for LLMs that combine linear and non-linear conversation representations.

CCS Concepts

• **Human-centered computing** → **Graphical user interfaces; Natural language interfaces; Interactive systems and tools.**

Keywords

Conversational interfaces, LLMs, Conversation graphs, Non-linear interaction, Branching interaction, Interactive systems

1 Introduction

Large language models (LLMs) have become integral to knowledge work, supporting activities such as writing, programming, planning, design, and research through iterative interaction [14]. During these tasks, users’ goals and interpretations often emerge progressively

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through dialogue with AI rather than being fully specified upfront, making LLM interaction a process of intentmaking and sensemaking [6]. As these interactions unfold, conversations accumulate alternatives, constraints, and reusable strategies, becoming external representations of users' evolving thought processes [2, 38]. Yet LLM interaction remains dominated by linear conversational interfaces that become difficult to manage as conversations grow longer and more exploratory. Earlier ideas and context become buried, while exploring alternatives often requires copying prompts, rerunning conversations, or distributing work across multiple chats or tabs [11, 19, 25, 31]. As a result, users must increasingly manage not only the task itself, but also the accumulated context, alternatives, and progress of the interaction [33].

Researchers have explored richer interaction paradigms for addressing these limitations, including graphical representations of generated responses [11], executable prompting graphs [3, 35], integrated rewriting environments [25], spatial canvases for organizing generated information [2, 30, 31], and conversation-level branching [15]. These systems demonstrate the value of introducing structure into LLM interaction, but often require users to explicitly construct or manage that structure by assembling prompt graphs, arranging workspace artifacts, or organizing conversational branches. This leaves open how conversational structure can emerge directly from ongoing interaction while remaining integrated with the familiar chat experience. To ground this problem in users' existing practices, we conducted a formative study with eight experienced LLM users. The study identified recurring challenges around context fragmentation, iteration friction, and exploring alternative conversational paths, motivating the design of CanvasConvo.

Guided by these findings, we developed CanvasConvo, a conversational interface that combines conventional chat with a synchronized spatial representation automatically derived from the evolving conversation. Every message becomes part of a persistent conversational structure that users can revisit, branch, and organize without separately authoring a workspace or abandoning the familiar interaction model of chat. Rather than positioning the spatial canvas as a replacement for chat, CanvasConvo explores how linear and non-linear representations can coexist as complementary views of the same evolving conversation. Because the system is intended to support conversations that accumulate, branch, and are revisited over time, we sought to understand how these representations become incorporated into users' ongoing practices rather than evaluating only their initial use. Prior work shows that the qualities shaping initial encounters with interactive systems can differ from those that become important through continued use [12], while longitudinal studies of generative AI have identified processes such as familiarization, customization, and workflow appropriation that emerge through repeated interaction [17]. We therefore conducted a five-day exploratory field study with 24 participants who incorporated CanvasConvo into self-directed writing, programming, planning, research, and study activities. Using interaction logs, questionnaires, and semi-structured interviews, we examined how participants adopted and appropriated its conversational representations over time.

Our findings reveal a division of labor between the two synchronized representations: participants primarily treated chat as a generation surface to advance the conversation, while using the

spatial canvas as an orientation surface to revisit, compare, and organize evolving conversational trajectories. Branching supported exploration of alternatives while preserving their relationship to the prior context, but also introduced new challenges in understanding which context a branch retained and in reconciling divergent conversational paths. Together, these findings suggest that supporting non-linear LLM interaction is not simply a matter of replacing linear histories with richer visual structures. Instead, conversational interfaces can benefit from treating linear and spatial representations as complementary interaction surfaces, supporting the fluid presentation of conversational content while making the accumulating structure of that interaction available for navigation, reflection, and exploration over time.

2 Related Work

We review prior work from three complementary perspectives: research on intent formation during conversational interaction with LLMs, approaches that structure LLM interactions beyond chronological chat, and spatial workspaces for supporting non-linear exploration. Together, these strands motivate our investigation of conversation-level spatial representations.

2.1 Intent Formation in Conversational LLM Interaction

Prior work characterizes interaction with LLMs as an iterative process in which users progressively formulate, evaluate, and refine their intent. Why Johnny Can't Prompt shows that non-expert users often develop prompts opportunistically, struggle to articulate their goals precisely, and lack systematic strategies for evaluating model outputs [39]. Similarly, research on intentmaking argues that goals and interpretations emerge through interaction rather than being fully specified in advance [6].

Several systems support this iterative process by externalizing intermediate conversational artifacts. Wordcraft enables users to iteratively rewrite, continue, and discuss generated text during collaborative writing [38]. PromptCanvas represents prompts and intermediate outputs as persistent, composable widgets that users can refine throughout creative workflows [2]. Likewise, Selenite structures generated alternatives and evaluation criteria to support reasoning during exploratory decision making [16]. Although these systems address different domains, they collectively show that intermediate prompts, outputs, and decisions become valuable artifacts that support the gradual refinement of user intent.

2.2 Structuring LLM Interaction

Beyond supporting iterative intent formation, researchers have explored alternative representations for organizing interactions with LLMs. Existing systems structure different aspects of the interaction process, including generated responses, prompting workflows, conversational memory, and information spaces.

Some systems focus on structuring model outputs. Graphologue transforms generated responses into node-link graphs of entities and relationships, enabling non-linear exploration of generated content [11]. VISAR represents argument development as an editable hierarchical tree, allowing users to organize and revise reasoning

across multiple levels of abstraction [40]. Other work instead structures prompting workflows. PromptChainer and AI Chains decompose complex tasks into modular prompt pipelines with intermediate outputs that improve transparency and debugging [35, 36], while ChainForge represents prompt workflows as executable graphs for systematic comparison and evaluation [3]. DirectGPT maps direct-manipulation operations to reusable prompts [19], and Memolet externalizes conversational memory into reusable interactive objects [37].

A complementary line of work structures broader information spaces assembled during exploration. Sensecape and Luminare organize generated information within hierarchical workspaces that support multilevel exploration and synthesis instead of committing users to a single conversational trajectory [30, 31]. Collectively, these systems demonstrate that explicit representations improve navigation, comparison, and reuse during complex interaction with LLMs, although they organize different artifacts of interaction rather than the conversation itself.

2.3 Spatial Workspaces and Non-Linear Interaction

Spatial workspaces have become an increasingly common interaction paradigm for supporting complex cognitive tasks. Earlier research on conversation visualization showed that asynchronous discussions naturally evolve into multiple interleaved threads and benefit from persistent spatial representations that preserve context while enabling flexible navigation [23, 24]. More recent work extends these ideas to interaction with LLMs.

PromptCanvas enables users to arrange prompts and generated outputs within a persistent workspace that supports iterative refinement through direct manipulation [2]. Sensemaking systems such as CoNotate, InterWeave, and ForSense integrate adaptive guidance into user-created workspaces to support synthesis across evolving information structures [20, 21, 24]. DataTone similarly demonstrates how interactions within a workspace can be transformed into persistent constraints that shape subsequent analysis [8]. Recent systems have also explored branching conversational interaction. Branchat explicitly preserves alternative conversational trajectories, allowing users to revisit and continue previous lines of exploration without overwriting earlier work [15]. Together, these systems demonstrate the value of spatial organization and non-linear navigation for exploratory interaction with LLMs.

2.4 Synthesis and Positioning

Prior systems move beyond linear chat by structuring different parts of LLM interaction in different ways. As summarized in Table 1, these representations range from generated response content and prompt workflows to information spaces and conversation histories. They also differ in how structure is created and how it relates to chat: some require users to construct workflows or organize workspace artifacts, while others derive structure from the interaction itself; similarly, structured representations may replace, augment, or remain synchronized with conversational interaction. Together, these systems demonstrate multiple ways of externalizing structure to support exploration, navigation, and revisitation.

CanvasConvo focuses specifically on the conversation history as the object of representation. Its spatial structure is derived from the evolving conversation, with users creating alternative trajectories through branching, and remains synchronized with the familiar linear chat interface. Rather than replacing chat with a separate visual workspace, CanvasConvo treats linear and spatial representations as complementary views of the same conversation. While prior work has established the value of structured representations, comparatively less is known about how users incorporate such conversation-level representations into exploratory and longer-running LLM workflows in practice.

Building on this gap, we examine both how users currently manage conversational complexity in existing LLM interfaces and how they subsequently interact with CanvasConvo during self-directed work. We focus on how users manage alternative directions, revisit and reuse prior conversational artifacts, and navigate non-linear structures during longer-running and multi-step tasks. We address the following research questions:

- RQ1** How do users currently manage multiple ideas, alternative directions, or *what-if* scenarios when interacting with LLM chat interfaces?
- RQ2** How does providing non-linear conversational structures (e.g., branching, visual navigation, and reuse of conversational artifacts) influence users' ability to revisit, reuse, and adapt prior prompts and responses?
- RQ3** How do users interact with such non-linear conversational interfaces when working on longer-running or multi-step tasks with LLMs?

3 Formative Study

Prior work has identified challenges in managing increasingly long and complex interactions with LLMs, including difficulties in revisiting earlier content, exploring alternative directions, and maintaining context across conversational turns. However, less is understood about how users encounter and work around these challenges in their existing LLM workflows, particularly when conversations involve iterative exploration rather than a single linear progression.

To better understand these practices and inform the design of CanvasConvo, we conducted a formative study examining how users manage complex and exploratory interactions with current conversational LLM interfaces. The study focused on participants' existing practices: how they explore alternative directions, iterate on previous outputs, maintain context, and organize conversational histories as their interactions grow.

Eight expert LLM users (F1-F8), all with prior experience using conversational AI tools, were recruited through purposive sampling under the inclusion criterion of intensive professional LLM use (10+ sessions/week, at least 6 months of regular use); three were UI/UX researchers and five were software developers. Participants took part through in-person sessions, remote sessions, or asynchronous written feedback, depending on their availability covering workflow walkthroughs, breakdown moments, ideal-tool sketches, and reactions to early design directions. Working through the sessions together, the two researchers grouped the recurring observations into three core themes.

Table 1: Comparison of representative LLM interaction systems by primary representation, structure creation, use case, relationship to chat, and empirical evaluation.

System	Primary Representation	Structure Creation	Primary Use Case	Relationship to Chat	Evaluation
Graphologue [11]	LLM response content	Automatic	Information exploration	Synchronized diagram and text	Qualitative study
ChainForge [3]	Prompt workflow	User-authored	Prompt engineering / testing	Separate visual workflow	Lab study
PromptCanvas [2]	Prompt logic	Hybrid	Creative writing	Canvas workspace	2 lab studies + field study
Sensecape [31]	Information space	Hybrid	Exploration / sensemaking	Structured workspace	Lab study
Branchat [15]	Conversation history	Automatic + user branching	Revisitation / context management	Synchronized tree and chat	Lab study
CanvasConvo	Conversation history	Automatic + user branching	Exploratory and long-running LLM workflows	Synchronized complementary canvas and chat views	5-day field study

3.1 Findings

Participants expressed consistent frustration with the linear, sequential nature of current LLM interfaces. Despite their expertise, many lacked the vocabulary to articulate what they wanted until the session prompted structured reflection. **Three themes were identified as central issues. Context fragmentation, iteration friction, and the forking problem.** These themes align with research on information foraging [22] and the tension between linear interaction paradigms and the non-linear nature of knowledge work [31].

3.1.1 Context Fragmentation. Context fragmentation refers to the scattering of relevant information across the conversation history, forcing users to scroll, search, or mentally reconstruct the state of their work. Participants consistently reported difficulty locating files and outputs, with F3 noting, “I spend half my time scrolling up to find the file I just uploaded. By the time I find it, I have lost my train of thought.” Valuable outputs such as well-structured code snippets or polished paragraphs became embedded in the scroll history and were effectively inaccessible for later retrieval. F6 illustrated the resulting inefficiency: “I got this perfect function like 20 messages ago, and now I cannot find it. I have to scroll and scroll, and sometimes I just regenerate it, which is wasteful.” Several participants also described the cognitive friction of session interruption. Upon returning after a break, they reported needing to rebuild their mental model of the entire conversation: “If I close the tab and come back an hour later, I have to re-read the whole thing to remember where I was” (F1). This behavior reflects the extraneous cognitive load imposed by interfaces that provide neither persistent summaries nor spatial overviews of session state [32].

Six of eight participants had developed explicit workarounds: maintaining separate documents (Notion, Word, Google Docs) to preserve important outputs from the chat. This coping strategy echoes the principles of casual information organization, in which users create ad hoc collections of resources when the primary interface fails to support the necessary persistence of artifacts [18]. The widespread adoption of this workaround reveals a clear gap: critical information that should be persistent and retrievable must instead be manually externalized by the user. The design implication is a persistent, always-visible artifact repository that elevates uploaded files and generated outputs from embedded ephemera to first-class, directly accessible objects.

3.1.2 Iteration Friction. Iteration friction refers to the disproportionate cost of making small, localized refinements to generated content [34]. Six of the eight participants described frustration with the binary choice between accepting an output as-is or regenerating it entirely. F2 articulated this constraint clearly: “I just want to tweak this one paragraph, not regenerate the whole essay. But if I ask it to fix paragraph three, it gives me a completely different version of the whole thing.” To circumvent this limitation, users developed complex workarounds. F5 described constructing increasingly unwieldy prompts designed to protect desired portions while modifying others: “I end up writing these enormous prompts like ‘keep paragraphs 1, 2, and 4 exactly the same, but rewrite paragraph 3 to be more formal.’ It is exhausting.” More concerning was the inhibitory effect on iteration itself. F7 reported a reluctance to ask for revisions even when dissatisfied: “Sometimes I get something 90% right, and I am scared to ask for changes because it might ruin what is already working.” This fear-driven acceptance of suboptimal outputs represents a fundamental usability failure.

Seven of the eight participants reported manually editing LLM outputs rather than requesting revisions, and F3 described an extreme adaptation: copying individual paragraphs into entirely new chats for focused iteration, then manually recombining the results across sessions. These strategies highlight a critical design need: inline tools for localized iteration that allow users to target specific content segments for refinement without triggering full-context re-prompting or risking unintended side effects elsewhere in the output. This aligns with the research calling for a shift from purely conversational interaction toward direct manipulation of specific generated objects [19]. F2 raised an important concern about transparency: “How do I know the widget is only changing the selected text? What if it’s also silently modifying the context around it?” This does not invalidate the case for local iteration tools; rather, it identifies a requirement that must accompany them (preview affordances such as diff visualization or undo-group mechanisms) to calibrate user trust [1].

3.1.3 The Forking Problem. The forking problem refers to the difficulty of exploring alternative approaches without losing accumulated context or polluting the conversation history. Seven of the eight participants encountered this tension between the desire to try new strategies and the cost of restarting from scratch. F8

quantified the friction: “If I want to try a different prompt strategy, I have to copy-paste everything into a new chat. That is like a five-minute tax every time I want to explore an alternative.” Some users attempted to work around this by continuing in the same chat, but this strategy traded technical convenience for cognitive overhead. F6 described the result: “My chat looks like a graveyard of half-finished ideas. I cannot tell which approach I am currently on.” Other participants adopted the multi-window strategy, opening parallel sessions to preserve context. However, this approach introduced new friction in the comparison phase. F1 reported: “I have to open two browser tabs side-by-side and manually compare. There is no way to see the branches together.” These ad-hoc workarounds (mental models of scattered branches, copy-paste overhead, manual cross-tab comparison) point to a fundamental absence: integrated support for exploring and comparing parallel lines of inquiry within a single coherent interface, a capability essential for sensemaking with LLMs [11, 31].

The severity of this problem manifested in different ways across the sample. F4 reported habitually maintaining “15 to 20 ChatGPT tabs open at any time,” each representing a distinct exploration thread. F2 adopted a more self-aware coping mechanism, typing explicit “BRANCH:” markers directly into the chat to impose conceptual structure: a fragile strategy that offloads to working memory a distinction that the interface should maintain automatically.

These challenges point to a spatial canvas with integrated branching support. The cognitive benefits of stable spatial layouts for information retrieval and navigation are well-documented [26], and recent work shows that spatial organization can substantially reduce the overhead of managing complex, multi-threaded explorations [27]. The design solution is to make branches visually salient, spatially distinct, and directly comparable.

This enthusiasm for spatial organization was not universal. F7 articulated a limitation: “I already have spatial organization in my editor. If I open yet another spatial interface for the LLM, am I coordinating two spatial models in my head, or does one replace the other? That feels complicated.” Spatial interfaces may thus introduce friction when layered on existing spatially complex tools such as IDEs; their value is contingent on task complexity, workflow integration, and the user’s broader tool ecosystem, a boundary condition the prototype accommodates through an optional embedded panel mode.

3.2 Design Goals

Our formative study revealed that users were already constructing implicit conversational branches through multiple chats, browser tabs, copy-pasting, and self-authored markers, creating a trade-off between preserving conversational context and keeping alternative trajectories manageable. Together with recurring challenges in retrieving and iterating on prior content, these observations, alongside prior work on non-linear LLM interaction, direct manipulation, and information organization, informed four design goals for supporting exploratory and long-running conversational workflows.

DG1 Support parallel exploration of conversational paths.

Prior work shows that users iteratively refine prompts and pursue multiple directions during creative and analytical

tasks [4, 39]. However, conventional chat interfaces constrain interaction to a single linear trajectory. Grounded in the *Forking Problem* identified in our formative study, the system should allow users to pursue alternative directions from existing conversational states without duplicating context or fragmenting their work across separate chats. These alternatives should remain connected to their shared conversational history so that users can move between and compare different lines of exploration.

DG2 Enable direct manipulation of conversational artifacts.

Informed by the *Iteration Friction* observed in our formative study, the system should enable users to act on specific conversational elements rather than relying exclusively on sequential prompting and full-response regeneration. Following principles of direct manipulation, conversational artifacts should remain persistent and operable [19], allowing users to selectively reuse, branch from, compare, or transform prior content.

DG3 Support scalable navigation and context management.

Grounded in the *Context Fragmentation* identified in our formative study, the system should help users maintain awareness of conversation structure as interactions accumulate. Users should be able to move between an overview of their conversational history and focused interaction following Shneiderman’s mantra [29] with particular branches, messages, or artifacts, supporting both orientation and retrieval in increasingly complex conversations.

DG4 Balance structured guidance with flexible interaction.

Throughout the formative study, participants developed different strategies for managing complex LLM interactions, including using external documents, multiple chat windows, and self-authored markers to organize branches. Structured interaction techniques, such as prompt suggestions and scaffolding, can support users in exploring and refining ideas [39]. At the same time, overly rigid systems may constrain user agency. The system should provide context-aware guidance while preserving flexibility in how users explore, structure, and manipulate conversations.

4 CanvasConvo

Building on our design goals, CanvasConvo conceptualizes interaction with LLMs as a *non-linear workflow* rather than a sequence of isolated turns.

4.1 Interaction Concept: Structuring Non-Linear LLM Workflows

Inspired by interaction paradigms from visual editing systems and prior work on structured writing environments, CanvasConvo combines a familiar chat interface with spatial and structural interaction mechanisms. The system organizes interaction across three complementary dimensions: spatial organization, branching, and temporal navigation, allowing users to fluidly move between focused dialogue and exploratory workflows.

4.1.1 Spatial Canvas. The spatial canvas serves as a structural representation of the conversation, transforming the linear message

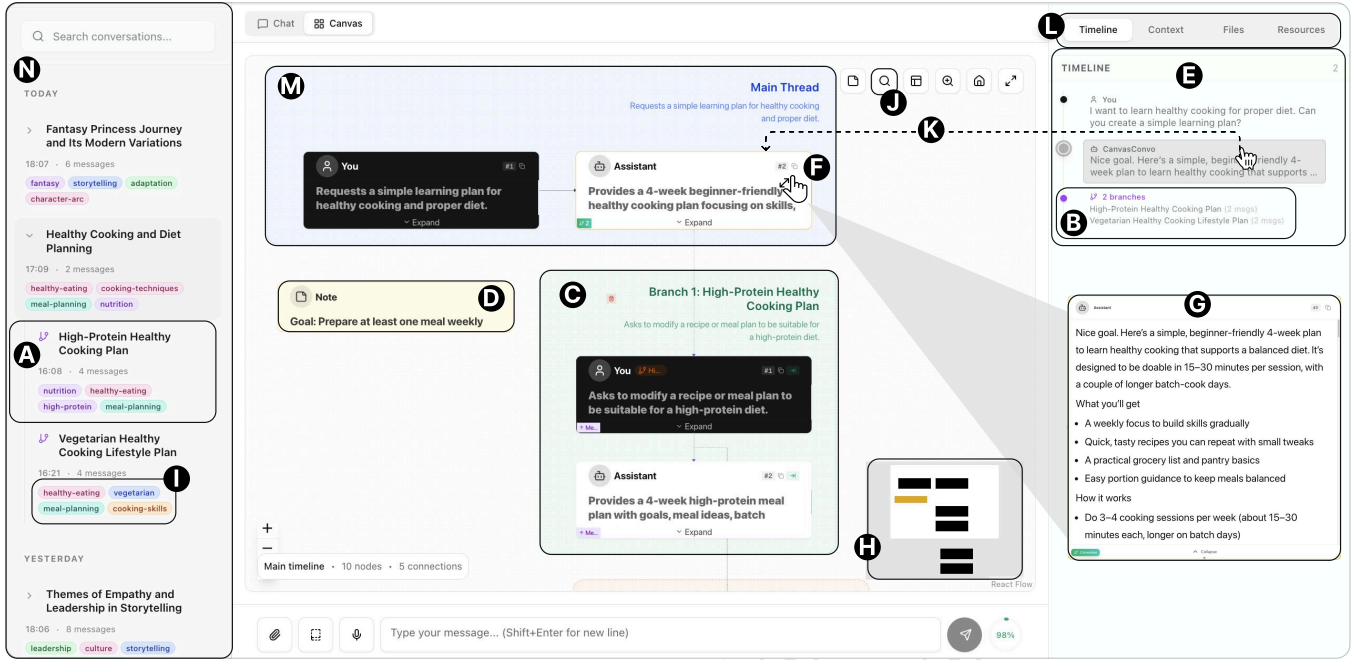


Figure 2: CanvasConvo (canvas view). The canvas (center) visualizes messages as nodes with a linear main thread (M), and branching zones (C, F), while the sidebar (N) organizes conversations (with tags (I)) and branches (A). Users can annotate with notes (D), navigate via semantic zoom (G) and a minimap (H), and search (J) to highlight relevant content. The timeline (E) shows conversation history and enables navigation to messages (K), with branch points explicitly marked (B). The overview panel (L) displays contextual information, including goals and resources.

sequence into a node-link graph. Each message is represented as a node, and edges encode sequential relationships and branch connections. The layout is **system-derived and structure-driven**. The main conversation thread is arranged linearly (Main Thread, Figure 2-M), while branches appear as separate zones connected to their origin points. The layout reflects the underlying conversational structure. The canvas introduces several key interactions:

- **Node Types.** Messages are represented as chat nodes containing role labels and content previews (Figure 2-F). Branches are grouped into *branch zone nodes*, visually distinct regions with labels and automatically generated summaries (Figure 2-C). Additional node types include loading placeholders and user-created sticky-notes for annotations (Figure 2-D). With these, users can add annotations, for example, to externalize ideas, without modifying the conversation.
- **Semantic Zoom.** The canvas supports four levels of semantic zoom (Cluster, Minimal, Compact, Full), from high-level role indicators to full message content (Figure 2-G). This enables users to move between overview and detailed inspection.
- **Navigation and Search.** Users navigate via panning, zooming, keyboard shortcuts, and a minimap (Figure 2-H) that provides a global overview. A text search function (Figure 2-J) highlights relevant nodes by adjusting opacity, allowing quick identification of content in large conversations. Hovering over any node in the canvas further highlights the corresponding conversational path.

Overall, the canvas functions as a *visualization and navigation layer* that makes conversational structure, relationships, and alternative paths explicitly visible (DG3).

4.1.2 Conversation Branching. CanvasConvo introduces branching as a core mechanism for exploring alternative conversational trajectories. From any message, users can create a branch that duplicates the conversation history up to that point and continues as a new thread (Figure 3-C). This preserves prior context while enabling divergence. Branch creation is integrated directly into the interaction flow via buttons in chat messages and canvas nodes. Branches are represented consistently across the views:

- as nested entries in the conversation sidebar (Figure 2-A)
- as badges attached to branching points in the chat (Figure 3-A)
- and as distinct zones in the spatial canvas (Figure 2-C)

Hovering over a branch tag reveals a preview of the branch (Figure 3-B). Branching supports workflows such as exploring alternative prompts, comparing different outputs, or recovering from earlier points in the conversation. By externalizing alternative paths, branching transforms interaction from a single linear progression into a *multi-path exploration process* (DG1). It is also possible to delete branches if needed.

4.1.3 History Timeline. To complement the structural view provided by the canvas, CanvasConvo includes a timeline view within the overview panel. It presents a vertical history of all messages, with each entry represented by a colored marker and a text preview

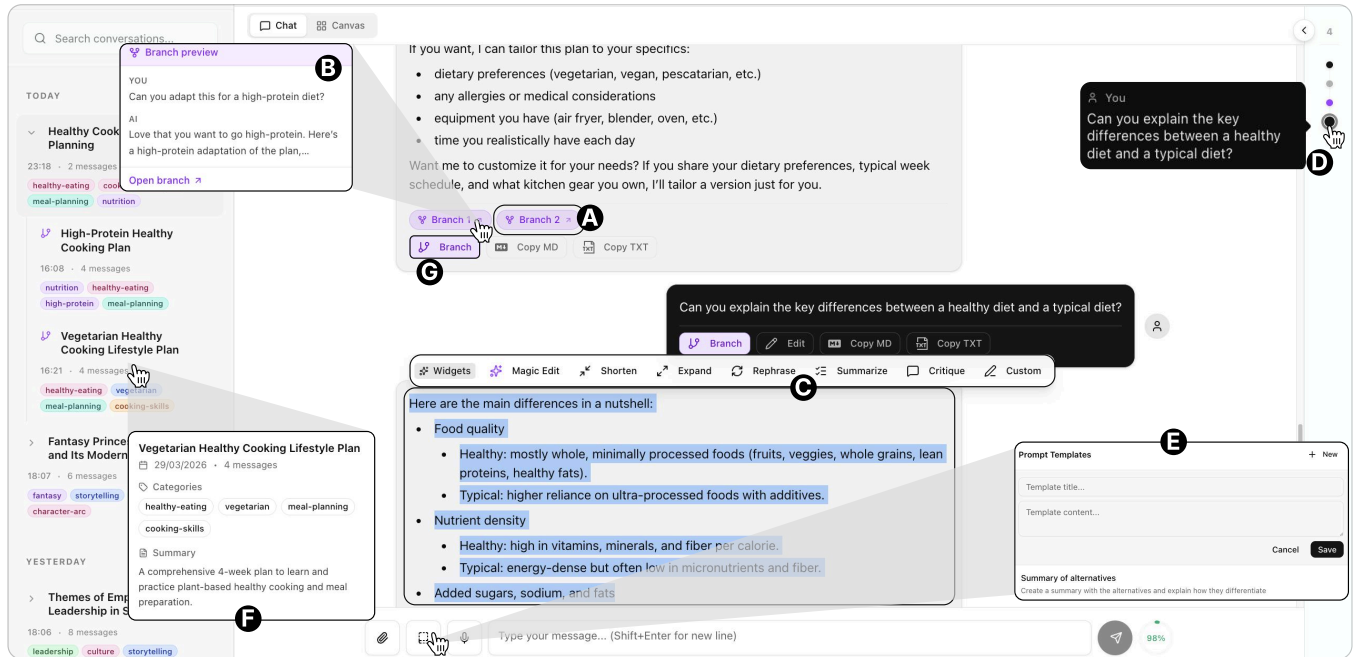


Figure 3: CanvasConvo (Chat view). Messages can be branched directly (A), with previews shown on hover (B) and branch points indicated in the thread (G). Hovering over a point in the timeline shows an overview of the conversation (D). A contextual toolbar (C) enables inline transformations such as summarizing or rephrasing selected text. The sidebar provides conversation overviews and metadata (F), while prompt templates (E) support reusable inputs.

(Figure 2-E). Selecting an entry navigates the viewport to the corresponding message in both chat and canvas view (Figure 2-K). Branch-point messages are explicitly indicated (Figure 2-B). Hovering over a timeline entry reveals a preview of the corresponding conversation text (Figure 3-D, there shown in the collapsed state of the timeline sidebar). This view emphasizes **temporal progression** (DG3), allowing users to:

- revisit earlier states of the conversation
- track how ideas evolved over time
- and quickly navigate across different points in the interaction

Together with the canvas, the timeline provides a complementary perspective: while the canvas shows *structure and relationships*, the timeline reveals *sequence and history* by preserving the chronological flow of both user and assistant messages.

4.1.4 Supporting Interaction Mechanisms. In addition to these core components, CanvasConvo provides several mechanisms that support interaction across views and with chat messages in general:

- **Chat Interface.** This view shows messages with real-time streaming responses and rich rendering (Markdown, code, LaTeX, images). Messages support actions such as edit, regenerate, copy, and branch. The sidebar on the left (Figure 2-N) shows a searchable list of conversations by recency; each displays metadata such as message count and conversation tags (Figure 2-O). The tags are generated based on the whole context of the conversation and dynamically updates as the conversation progresses. Hovering reveals a summary preview (Figure 3-F). At the bottom, the

message input supports reusable prompts via a picker (DG4), to quickly insert and reuse templates (Figure 3-E).

- **Toolbar.** Selecting text within a message reveals a toolbar with transformation actions (e.g., shorten, expand, rephrase, summarize, critique), see Figure 3-C. When clicking one, a popup shows a preview for users to confirm. These tools enable localized refinement of outputs without writing new prompts (DG2).
- **Context and Resource Management.** The overview panel provides access to contextual controls (e.g., inferred goals, system instructions), uploaded files, and automatically extracted resources such as links and generated images (Figure 2-L), collapsed in Figure 2. This makes otherwise implicit context visible and configurable.

4.2 Technical Implementation

CanvasConvo is implemented as a single-page web application using Next.js (App Router) and React, with Supabase providing authentication, persistence, and real-time synchronization. The spatial canvas is built on React Flow. Model responses are streamed via Server-Sent Events to enable low-latency interaction. The system integrates multiple LLMs (e.g., GPT-5, GPT-4.1, GPT-4o), with configurable reasoning effort and large context windows supporting extended interactions.

5 Main Study Design

We conducted an exploratory study to investigate how users interact with our system. We were interested in understanding how users managed long-running LLM-supported workflows as they

incorporated CanvasConvo into their everyday tasks. Participants were free to use the system for their own writing, programming, planning, research, study, and other knowledge-work activities, allowing us to observe how conversation-level spatial representations were appropriated across naturally occurring workflows. The study was approved by our institution's ethics board.

5.1 Participants

We recruited 24 participants (P1-P24, with no overlap with participants from the formative study) through institutional mailing lists and word-of-mouth (6 female, 17 male, 1 non-binary) with ages 21 to 54 years ($M = 28.6$, $SD = 6.6$). They represented diverse professional backgrounds, including students, researchers, software engineers, and industry professionals. Participants varied in their writing experience, including both professional and non-professional writers. All reported prior experience with LLM-based tools, with most indicating high familiarity and frequent use in their daily workflows. Participants commonly used LLM systems for tasks such as writing and editing, coding, data analysis, brainstorming, planning, and learning. Overall, the participant pool reflects a group of active LLM users with varied domains of application and experience levels. A detailed overview is provided in Table 2. Participants were compensated 20€ for the five-day study.

5.2 Study Procedure

Our study allowed participants to integrate the system into everyday workflows. The procedure combined a structured onboarding task with open-ended usage.

5.2.1 Pre-Task Preparation. Before starting the study, participants completed an introductory questionnaire, watched a tutorial video¹ introducing the system, and got access to CanvasConvo. The questionnaire gathered data on participants' experiences with multi-step workflows in LLM chats, including how they structure complex tasks, manage multiple lines of thought, reuse prompts, and organize conversations. We further assessed perceived challenges, satisfaction with current interfaces, and desired features for supporting parallel exploration and long-running tasks.

5.2.2 Guided Mandatory Task. Participants first completed a structured, mandatory task during the five-day study period. They selected or defined a topic such as a learning plan, weekly routine, creative writing idea, or personal project. They created an initial version of their idea through interaction with the AI, iteratively refined it via follow-up prompts, and explored alternative versions by modifying aspects such as difficulty, constraints, or style. Participants were encouraged to create multiple branches, reuse prior prompts, revisit earlier responses, and use the canvas view to understand the conversation structures.

5.2.3 Open-Ended Usage. After completing the guided task, participants used the system freely for their own work during the remainder of the five-day study. These activities included planning, programming, research, ideation, and other self-directed LLM-supported workflows, allowing us to observe adoption of the interface under naturalistic usage conditions.

¹submitted as a supporting material to this submission

5.2.4 Post-Task Measures and Interview. At the end of the study, participants completed a post-task questionnaire on usability, task load, and questions on workflow support and conversation management, and participated in a semi-structured interview. The interviews focused on participants' experiences, workflows, perceived benefits and challenges, and suggestions for improvement.

5.3 Measurements and Analysis

We collected both quantitative and qualitative data to capture user behavior and perceptions.

5.3.1 Quantitative Measures. Participants completed standard usability and workload instruments, including the System Usability Scale (SUS) [5] and NASA Task Load Index (NASA-TLX) [9] along with custom Likert items on workflow support, revisiting and reuse, managing multiple directions, and sense of control and authorship. We also logged interaction data, including branching behavior, node creation, navigation patterns, and reuse of conversational artifacts.

5.3.2 Qualitative Analysis. Interview recordings were transcribed and independently reviewed by two researchers. Because the interview protocol was organized around topics aligned with our research questions, including participants' existing conversational workflows, navigation and revisitation, exploration through branching, and experiences with the spatial representation, the analysis focused on identifying recurring patterns within and across these discussion areas. The researchers iteratively compared their observations, grouped related responses, and discussed emerging patterns and differences until reaching consensus. The resulting themes were used to organize the qualitative findings, with representative quotations used to illustrate both recurring experiences and divergent cases.

6 Results

We report the qualitative findings first, beginning with participants' existing practices in conventional LLM interfaces and then examining how they adopted, adapted, and encountered challenges with CanvasConvo during the five-day deployment. We subsequently relate these statements to patterns observed in the interaction telemetry.

6.1 Practices and Challenges in Conversational UI Workflows (RQ1)

Our findings from the introductory questionnaire (*before seeing/using our design, separate from the formative study*) reveal how participants currently structure, manage, and struggle with workflows in their usual LLM chat interfaces.

6.1.1 Structuring Multi-Step Tasks. Participants described breaking complex tasks into smaller steps and iterating incrementally. They followed structured, step-by-step approaches, often starting with a high-level plan before refining individual components: *"I usually ask one chat for a plan of the complex task... and then start new chats for each task."*—P15. Others described iterative refinement within a single thread: *"I outline the text... draft with AI... give feedback... iterate."*—P1. Several participants emphasized decomposition and

incremental problem solving: *“I go step by step, with a bottom-up approach... solving a smaller problem first.”*—P5. However, not all workflows were structured. Some participants reported more organic and less controlled approaches: *“I usually start chatting and work through the task organically... my workflow can sometimes feel unstructured”*—P17.

6.1.2 Managing Conversations: One Thread vs. Multiple Chats. Participants described their typical use of LLM chat interfaces as relying on a single, continuously growing conversation thread, which often became difficult to manage over time. For example, P7 noted that they usually work within *“one long thread.”* As conversations grew, this structure led to a loss of clarity and increasing complexity. P2 described how they would *“fill it with so much unnecessary paths”* and eventually need to restart the chat altogether. Others characterized this experience as disorienting, with P14 noting that one can *“go down a rabbit hole... and not really know how you got there.”* To cope with these limitations, participants reported switching between multiple chats or restarting conversations entirely, leading to fragmented workflows and loss of context. Some split tasks across multiple chats to improve clarity: *“I split them... to have a better overview of the ‘big task.’”*—P16. A common strategy combined both approaches. For example P10 noted, *“If the topic can be split... I split them up in multiple chats.”* This highlights a tension between context continuity and organizational clarity, with no single strategy fully addressing both.

6.1.3 Managing Multiple Ideas and Exploration. When exploring alternative directions, participants relied on ad-hoc strategies such as scrolling, copying content, or opening multiple chats or browser tabs: *“Continue in the same chat and scroll back... copy prompts into another chat”*—P11. Some used external tools to compensate for missing functionality: *“I usually write external notes... to avoid unnecessary context.”*—P15. Despite these efforts, participants reported limited support for parallel exploration: *“One thread can’t always handle different tasks... I have to start another thread.”*—P7.

6.1.4 Loss of Overview in Long Conversations. A common challenge was maintaining an overview in longer interactions, due to the difficulty of retrieving earlier information or understanding past decisions. P8 said, *“It is hard to find previous answers or go back and build upon them.”* Similarly, P10 said, *“When the chat gets longer, it is difficult to keep track of what I discussed before.”* Participants attributed this to the linear, scroll-based nature of chat interfaces, such as P11: *“Scrolling a lot to previous chats.”*

6.1.5 Need for Structure, Navigation, and Branching. In their reflections on their usual tools, people consistently expressed the need for better organization and navigation, such as the ability to explore alternatives without losing prior context. P12 mentioned, *“I would add a way to create separate threads inside the same chat... to explore different ideas without mixing everything”*. P23 noted, *“There is no true ‘undo’ or branching feature... you’re often forced to start over”*. Other desired features included grouping, visual overviews, and improved navigation: *“A visual overview or map of the conversation”*—P8. P10 mentioned, *“Organizing the chat into different sections or topics with an overview”*.

Participants also highlighted the need for better retrieval and reuse: *“Being able to save certain prompts to use them afterwards.”*—P15. Overall, these findings point to limitations of linear chat UIs in multi-step and exploratory workflows, and thus confirm that our design features address a user need.

6.2 Effects of Non-Linear Structures on Reuse, Navigation, and Exploration (RQ2)

We next report the qualitative findings from participants’ experiences with CanvasConvo, specifically on how the non-linear structures support revisiting, reuse, navigation, and exploration.

6.2.1 Adopting Spatial Interaction Alongside Linear Chat. Participants contrasted CanvasConvo with traditional chat interfaces. Participants described preferring traditional chat interfaces for quick, low-effort interactions. For example, P3 noted that *“for very minimal searches... I would use the other ones”*, while P1 stated *“for quick things... I would just use a normal chat”*. Some continued to rely on linear interaction habits (P19, P23): *“I just scrolled up... I never ended up using the canvas due to force of habits.”* (P19), others characterized their prior LLM use as sequential and difficult to organize in contrast, such as P17: *“I normally use LLM tools in a very linear way, just in a long chat.”*

These statements indicate that adopting the spatial representation was not immediate or uniform. For some participants, established chat-based practices remained the default even when alternative navigation mechanisms were available. In particular, P19’s continued reliance on scrolling suggests that introducing a spatial representation does not by itself displace interaction habits developed through conventional chat interfaces. Participants instead differed in when and for which kinds of tasks they considered the additional structure worthwhile.

In contrast, participants described CanvasConvo as supporting exploration and organization: P11 described it as *“the go-to solution when I’m learning something,”* and P1 highlighted its usefulness *“for longer ideations... trying out different topics... the branching is really nice.”* More broadly, participants pointed to its applicability in larger-scale tasks, as reflected by P23’s observation that *“people working on bigger projects would benefit the most.”* And P7 said, *“usually... I have to use one long thread... but here... the organization was better and time efficient.”* They adopted spatial metaphors, framing the system as a map or structured overview: *“like a mind map... where your ideas branch out.”* (P6). P3 perceived it as a tree, saying *“it’s kind of like a flow chart or like a tree”*.

Together, these findings suggest that participants did not simply replace linear chat with a spatial workspace. Rather, they selectively incorporated spatial interaction when conversational complexity made the additional structure useful, while continuing to rely on familiar linear interaction for shorter or simpler tasks.

6.2.2 Externalizing Conversation State and Supporting Navigation. Participants highlighted how the canvas reduces the need to rely on memory or scrolling, by visualizing branches and conversation states: *“You can just click on older conversations... it was very easy to navigate.”*—P11. P17 noted, *“I could clearly see where I had to go and how the conversation developed”*. Participants described this as enabling faster comprehension of prior work, such as P19: *“It*

gives you a much faster way to get a feeling of what has happened". P9 explicitly compared the interaction to human conversational behavior: "It's more like how we talk to people... you can go back to something you mentioned earlier and continue from there". Participants described the canvas as making prior conversational states directly accessible rather than requiring them to reconstruct context from a linear history. Instead of rereading long transcripts, they used the timeline to jump to earlier moments in the conversation and the spatial layout to distinguish alternative reasoning paths. As P10 explained, "There was a timeline on the right side... I could just hover over the timeline dots" to revisit previous parts of the conversation, while DP6 highlighted the ability to move fluidly between "the overview" and detailed content. These accounts suggest that the canvas supported navigation not simply by making conversations easier to browse, but by externalizing conversation state into a persistent spatial representation that reduced the need to mentally reconstruct where ideas originated and how they evolved.

6.2.3 Branching as a Mechanism for Exploration and Control. The central mechanism for non-linear exploration was branching, but participants did not always develop an accurate mental model of its semantics. P23, for example, explained: "I didn't know what the branch knows and what it doesn't know." This uncertainty concerned the conversational context inherited by a branch rather than the mechanics of creating one. It suggests that making alternative paths visually explicit is insufficient if the relationship between their underlying conversational contexts remains ambiguous.

Participants used branching to explore alternatives, separate ideas, and avoid interference between different lines of reasoning. P19 said, "I used the branching tool to explore different ideas and evaluate them". P7 used branching in a highly structured, task-oriented way, saying "I made one branch for... materials... another branch for... logistics, guest list, and to-do things.". Participants also emphasized control and independence between branches. P10 explained, "The branching kind of gave me an incentive to explore more directions." These findings show a tension in branching interactions: branches provided participants with explicit spaces for exploring alternatives, but their usefulness depended on participants' understanding of which conversational states were shared or isolated across those spaces.

6.3 Interaction with Non-Linear Interfaces in Multi-Step and Long-Running Tasks (RQ3)

In the open-ended part of the study, participants applied Canvas-Convo to a variety of tasks over several days, including planning, programming, research, and ideation. Insights from these experiences extend and substantiate the previous aspects with concrete instances across real tasks.

6.3.1 Supporting Parallel and Iterative Workflows. Participants described using branching to explore alternatives in parallel and structure their work across multiple directions. Rather than progressing sequentially, they distributed related ideas across branches and worked on them simultaneously. As P5 explained, "Well, I'm also looking at something else in parallel, so that way I speed up my work". Similarly, P6 described organizing different aspects of a task across branches: "Something like... day one this was the diet plan...

what was the exercise routine... different kind of simultaneously, but two different branches." This ability to externalize parallel lines of thought supported more structured and efficient exploration.

Branching was also used to systematically explore variations of the same idea through prompt reuse. Participants iterated on prior inputs while preserving earlier results. For example, P10 noted, "I reused the previous prompts... to create different branches... what if I had one hour per day... or 20 minutes per day?", while P5 described applying the same prompt across conceptual variations: "The same prompt we used, but with a different... concept... polymorphism... virtualization... singleton pattern...". This pattern enabled controlled experimentation across alternatives without losing context. Participants further revisited earlier states and resumed exploration from previously abandoned paths. As P7 explained, "I went to the Canvas option... went back in conversation where I abandoned another idea, and then... picked up the conversation from there again." This iterative navigation allowed users to treat conversations as revisitable states rather than linear histories. However, P20 noted, "Switching between canvas and chat was confusing". This difficulty highlights a second adoption challenge distinct from branch semantics. Although the synchronized views exposed the same underlying conversation, moving between linear and spatial representations required participants to maintain orientation across two different interaction models. For some participants, the benefit of obtaining a structural overview, therefore, came with an additional coordination cost when returning to a focused conversation.

Participants also noted limitations in comparing and integrating outcomes across branches. While the system supported divergence, synthesizing results remained challenging. As P3 suggested, "maybe merging the two ideas to see how would they look like". Overall, while branching enabled parallel exploration and iteration, it introduced new challenges around cross-branch comparison and integration.

6.3.2 Reflection and Interpreting the Spatial Representation. Participants reported that the system supported reflection and helped them better understand their own thinking processes. By externalizing conversation structure, the canvas enabled users to reason about their progress and decision-making. As P23 noted, "It helped me understand how to approach the task". Participants also used branching for implicit comparison, with P11 explaining, "You can compare outputs very effectively by switching branches". This suggests that the representation supports metacognitive engagement during task execution.

The spatial layout was frequently interpreted through familiar structural metaphors, particularly maps and organized overviews. P14 described the canvas as "like a map where I could see how the conversation developed", while P19 emphasized clarity and structure: "It was very structured and I was able to [get an] overview". Participants highlighted that the arrangement of branches made relationships between ideas easier to interpret. As P11 stated, "The sub-conversations are ordered perfectly... very overview-friendly".

In addition, the representation supported fluid transitions between overview and detail, enabling users to navigate different levels of abstraction. P6 explained, "you can see from the overview, and then you go into details", and noted that branching allows ideas to be considered "side by side... otherwise you go up and down... but here... it helps you think in an organized way."

Across these experiences, appropriation of the spatial representation was therefore uneven rather than universal. Some participants incorporated the canvas and branching into their workflows as mechanisms for organizing, revisiting, and comparing conversational paths, while others continued to rely on familiar linear interaction or encountered difficulties understanding branch context and coordinating between views. These differences suggest that adopting a non-linear conversational interface involved developing new interaction practices and mental models, rather than simply learning where interface features were located.

6.4 Subjective Evaluation

In this section we discuss participants' subjective experience of CanvasConvo using standardized usability and workload measures (SUS, NASA-TLX), along with Likert-scale ratings of agency, authorship, and system support. These measures capture how users perceived control, collaboration, and their ability to manage complex, multi-branch interactions.

6.4.1 System Usability Score. For the system usability score (SUS), CanvasConvo achieved a mean score of 78.9 ($SD = 11.2$), with a median of 80.0 and a range from 56.7 to 94.4. This places the system well above the standard usability benchmark of 68, indicating strong overall usability. The majority of ratings fall within the "good" to "excellent" range, with multiple participants assigning scores above 85, reflecting consistently positive perceptions of the system.

6.4.2 Perceived Workload (NASA-TLX). We analyzed raw NASA-TLX ratings for our tool to assess perceived workload. Overall, the results indicate low workload across all factors (see Table 3 in Appendix). Participants reported particularly low levels of physical demand ($M = 1.50, SD = 1.10$) and annoyance ($M = 1.67, SD = 0.82$), while mental demand ($M = 2.83, SD = 1.55$) and effort ($M = 3.29, SD = 1.68$) were slightly higher but still in a low-to-moderate range. The overall NASA-TLX score ($M = 2.79, SD = 0.74$) further suggests that participants experienced the system as lightweight and manageable.

6.4.3 Authorship and Agency. Participants reported a generally strong sense of agency and alignment when interacting with the tool, see Figure 4. Most participants agreed that they could easily guide the system (91%) and felt in control of the conversation and its outcomes (83%). Similarly, a large majority indicated that the outputs reflected their own ideas and intentions (87%), suggesting that the system supports user-driven expression rather than overriding it. While perceptions of ownership were slightly more mixed, a majority still felt that the ideas developed during the interaction belonged to them (65%). Participants largely experienced the interaction as collaborative (78%), indicating that the tool was seen not just as a passive assistant but as an active partner in the process.

6.4.4 Perceived System Support. Participants also reported strong support for managing and navigating longer, more complex interactions (see Figure 5). A large majority agreed that the system helped them keep track of ideas across extended conversations (91%) and maintain an overview of their work (96%). Nearly all participants found it easy to revisit earlier parts of the conversation (96%), indicating effective support for accessing prior context. Similarly, most

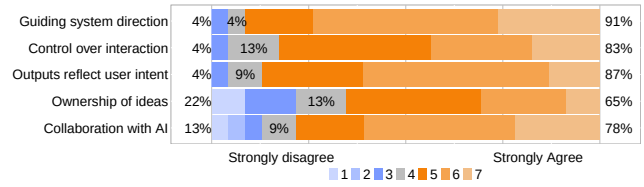


Figure 4: Participant ratings of perceived agency, authorship, and collaboration when using the system, measured on a 7-point Likert scale (N=24).

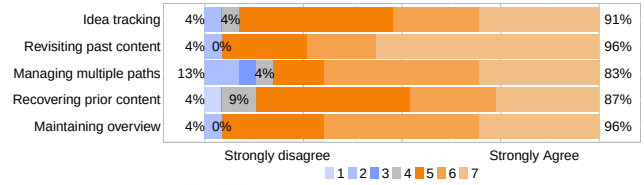


Figure 5: Participant ratings of the system's support for overview and context management across interaction, measured on a Likert scale (N=24).

participants reported that it was easy to recover previous prompts or results when needed (87%). While slightly more varied, responses still showed that participants felt supported in managing multiple directions or ideas within a conversation (83%).

Overall, participants rated the system positively for organization, navigation, and continuity in multi-step interactions. These aggregate ratings should, however, be interpreted alongside the qualitative findings, which show that participants differed in how readily they incorporated the spatial representation into their workflows and in how clearly they understood its interaction semantics.

6.5 Objective Telemetry

To complement participants' self-reported experiences, we analyzed interaction logs collected through a custom instrumentation layer during the field study. Across 24 participants, the system recorded 165 distinct conversations and 1,225 messages. On average, participants generated 51.04 messages and 6.88 conversations, indicating sustained engagement with the system over multiple sessions. Summary statistics are presented in Table 4 in the Appendix.

6.5.1 Feature Engagement and Interaction Distribution. Interaction activity was unevenly distributed across system components. Chat-based interactions accounted for the largest proportion of events (47.2%), followed by canvas interactions (28.3%) and sidebar navigation (11.4%). In addition, navigation behaviors such as branch revisits and timeline interactions were frequently observed, with 93 branch navigation events and 132 timeline clicks, highlighting active use of the system's mechanisms for revisiting prior context. Prompt templates were used 18 times in total, and 5 participants created templates.

The distribution, therefore, does not indicate a shift from chat-centered to canvas-centered interaction during the deployment. Instead, participants continued to perform most recorded interactions through the familiar chat interface while selectively engaging

with the spatial representation and its navigation mechanisms. This pattern is consistent with the qualitative findings in Section 6.2, and Section 6.3, where participants described the canvas as particularly useful for obtaining an overview, revisiting prior states, and managing alternative directions, but not necessarily as their primary surface for conversational interaction.

Prompt templates showed substantially narrower adoption: only five participants created templates during the deployment. This suggests that reusable prompting was not a central part of most participants' workflows during the study, despite being available as a supporting mechanism.

6.5.2 Temporal Patterns of Use. A comparison of recorded view durations shows that canvas-view events had a higher mean duration ($M = 34.1$ minutes) than chat-view events ($M = 13.0$ minutes). This difference should be interpreted cautiously. The telemetry records wall-clock time between entering and leaving a view and does not distinguish active interaction from reading, inactivity, or time spent in a background tab. Moreover, participants may have selected the canvas for more complex tasks, and lower familiarity with the spatial interface may itself have increased interaction time. The current data therefore cannot determine whether longer canvas durations reflect more sustained organization and reflection, greater task complexity, lower interface familiarity, or a combination of these factors.

In conjunction with the qualitative findings in Section 6.2, the duration pattern is consistent with participants using chat for relatively rapid input and iteration and the canvas for longer episodes of reviewing and organizing prior content. However, the telemetry alone does not establish these functional roles. A controlled within-participant study using equivalent tasks, active-time measurements, and task-level covariates would be needed to disentangle interface effects from differences in familiarity and task complexity.

6.5.3 Branching and Exploratory Behavior. Participants engaged with branching as a mechanism for creating alternative conversational trajectories. In total, 79 branches were created ($M = 3.29$ per participant), including 15 sub-branches and a maximum nesting depth of two levels. The telemetry also recorded 93 branch-open events ($M = 3.88$ per participant), indicating that participants navigated to existing branches in addition to creating them.

However, because the number of branch-open events only modestly exceeded the number of branch-creation events, these aggregate counts do not establish that branches were frequently revisited. The data also do not distinguish an immediate opening of a newly created branch from a later return to an earlier branch. We therefore interpret these events more conservatively as evidence that participants both created and navigated among branching conversation paths. Together with the qualitative findings in Section 6.2, this suggests that at least some participants used branching to structure alternatives as navigable conversational trees rather than solely as linear sequences.

6.5.4 Interaction with the Spatial Canvas. Telemetry from the canvas interface highlights active engagement with spatial navigation features. Zoom interactions were the most frequent action ($N = 881$; $M = 36.71$), followed by node selections ($N = 246$; $M = 10.25$) and search usage ($N = 94$; $M = 3.92$). The high frequency of zoom

interactions shows that participants frequently changed the level of detail displayed on the canvas. Together with participants' descriptions of moving between overview and detailed inspection, this pattern is consistent with the use of the canvas for navigating across levels of conversational detail. This aligns with the results discussed in Section 6.3. Temporal navigation was also actively used. Participants clicked on timeline entries 132 times ($M = 5.50$), directly navigating to earlier points in the conversation. This provides behavioral evidence supporting participants' reported ease of revisiting prior context (Section 6.2).

Overall, the telemetry data complement our qualitative findings by revealing distinct interaction patterns across the interface components. Chat interactions were more frequent, while canvas sessions tended to be longer and were accompanied by navigation actions such as branching, node selection, and timeline use. Although these behavioral traces do not by themselves establish how participants used each interface, they are consistent with interview accounts describing chat as supporting rapid conversational interaction and the canvas as facilitating organization and exploration during longer-running tasks.

7 Discussion & Future Work

In this work, we investigated how spatial and non-linear conversational interfaces can support LLM-based workflows. Specifically, we examined (RQ1) how users manage multiple ideas and alternative directions in traditional LLM chat interfaces, (RQ2) how non-linear structures influence reuse and navigation of conversational artifacts, and (RQ3) how users adopt and appropriate spatial conversational interfaces in longer-running tasks.

Limitations. One limitation of our study is the duration and scope of the field deployment. While the five-day study allowed us to observe evolving workflows and repeated interactions, longer-term use may reveal different patterns of adoption. Additionally, our participant pool consisted largely of experienced LLM users, which may limit generalizability to novice populations. Future work should investigate how such systems are adopted over extended periods and across different expertise levels. A further limitation concerns variability in participants' engagement and task characteristics. Participants worked on self-selected activities that differed considerably in duration, complexity, and information density, resulting in different cognitive and sensemaking demands across individuals. Finally, subjective measures such as NASA-TLX should be interpreted cautiously, as observed workload differences may partially reflect variation in underlying tasks rather than interface effects alone. Future studies could employ more controlled task sets alongside ecological deployments to better isolate the effects of interaction design.

Spatial Representations Complement Rather Than Replace Linear Chat. Our findings suggest that the transition from linear to spatial conversational interaction is better understood as appropriation than replacement. Participants continued to rely heavily on the familiar chat interface while selectively incorporating the canvas when they needed to revisit prior states, obtain an overview, or manage alternative directions. This division of labor is reflected both in participants' accounts and in the interaction telemetry, where

chat interactions remained more frequent than canvas interactions. Importantly, adoption of the spatial representation was uneven. P19's continued reliance on scrolling illustrates how established interaction habits can persist even when alternative navigation mechanisms are available. P20's difficulty switching between canvas and chat highlights the coordination cost introduced by maintaining synchronized linear and spatial representations, while P23's uncertainty about inherited branch context shows that users must also develop an accurate mental model of non-linear conversation semantics. Together, these findings suggest that designing non-linear LLM interfaces requires supporting transitions between representations and communicating conversational state, not simply exposing additional structure.

Externalizing Conversations as Navigable Structures. Our results show that representing conversations as spatial structures changes how users interact with LLMs. Instead of treating conversations as long transcripts, participants engaged with them as persistent, navigable artifacts, often describing the system as a “map” or “tree” of ideas. This aligns with prior work emphasizing the importance of external representations for sensemaking and exploration [2, 11]. Unlike such prior approaches that visualize single LLM responses [11, 40], CanvasConvo structures the entire interaction as a branching history, enabling parallel exploration and better workflow awareness. Features like spatial layout, timeline navigation, and semantic zoom supported smooth transitions between overview and detail [28], as also reflected in the interaction logs. Additionally, search, tagging, and automatic summaries helped users navigate and retrieve information in larger conversations.

Managing Cognitive Demands. People's use of our system and their reflections in comparison to their usual tools suggest that the linear chat paradigm is effective for short, goal-directed tasks but breaks down in exploratory and multi-step workflows. Concretely, for their usual LLM-based tools, they described difficulties navigating verbose outputs, maintaining context, and comparing alternatives, often resorting to external tools or fragmented strategies such as copying prompts or splitting tasks across chats. These observations align with prior work showing that LLM interaction introduces significant metacognitive demands, particularly in monitoring progress and evaluating outputs [7, 33]. Based on participants' experiences in our study, by externalizing conversational structure, CanvasConvo is perceived as reducing reliance on memory and scrolling. However, these new representations come with interaction demands. Future designs could explore further combinations, transitions, and adaptations of linear and non-linear conversation UIs.

Branching as a Mechanism for Exploration. Our branching feature was accepted and leveraged for exploring *what-if* scenarios. Unlike linear chats, it lets users explore alternatives without restarting, while preserving context. Participants used it to test ideas, compare approaches, and separate lines of reasoning, lowering the barrier to exploration [30]. Branching also supported users' sense of control, as branches were seen as independent spaces.

To compare results across branches, participants developed their own strategies, such as adopting reusable prompt templates to summarize and transfer insights. Since the conversation is structured

as a tree, it supported divergence but not merging. Future designs could address this gap by introducing mechanisms for merging and reconciling branches. For example, such designs could draw inspiration from merging in version control systems such as Git.

At the same time, our findings suggest that branching requires clearer interaction semantics than current implementations provide. While participants appreciated the ability to isolate ideas, several were uncertain about exactly what conversational state was preserved across branches. This ambiguity occasionally reduced confidence in the system despite branching operating correctly. These observations suggest that future systems should make context inheritance explicit—for example through visual indicators of shared history, inherited prompts, or branch-specific summaries—to better align users' mental models with system behavior.

Supporting Multi-Step and Long-Running Workflows. In longer-running own tasks, participants viewed CanvasConvo as a workspace rather than a chat tool, as it supported parallel exploration, iterative refinement, and revisiting prior states (with timeline navigation). This is consistent with work showing that externalizing cognitive work into manipulable representations supports more effective reasoning across complex, multi-step tasks [13]. Working with CanvasConvo thus supported planning, ideation, and problem-solving processes. Telemetry data further supports this distinction. Interestingly, participants naturally developed a division of labor between interaction modes. Linear chat supported rapid iteration and prompt refinement, whereas the spatial workspace was primarily used for organization, revisiting previous work, and managing multiple concurrent directions.

Authorship, Agency, and Control in Non-Linear Interaction. Participants reported a strong sense of agency when interacting with CanvasConvo. Prior work on human-AI co-writing has shown that close integration of AI raises questions of authorship, disclosure, and cognitive ownership [10, 38]. In our study, the introduction of branching supported a sense of ownership by making the model's context visible and controllable. This allowed users to isolate ideas, explore alternatives independently, and avoid unintended carry-over between interaction paths. Related, participants' perceived that outputs largely reflected their own intentions. The system was frequently perceived as a collaborative partner rather than a passive tool, indicating a shift toward co-creative interaction. These findings suggest that non-linear interfaces can strengthen perceived agency by identifying user decisions and interaction paths.

8 Conclusion

We presented CanvasConvo, a conversational interface that transforms linear LLM interaction into a non-linear, spatial workspace through branching, a canvas, and timeline-based navigation. In a 5-day field study with 24 participants, we observed how users structured multi-step workflows, explored alternatives in parallel, and revisited prior interactions within this space. Participants used branching to separate ideas and explore directions, while the canvas, timeline, and supporting features (e.g., summaries, tags, reusable prompts) helped maintain an overview and reuse conversational content. These findings show how conversation-level spatial representations can support exploration, navigation, and

sensemaking in LLM-based work, while also revealing that their adoption is neither immediate nor uniform. Participants selectively combined spatial interaction with familiar linear chat practices, and the added structure introduced challenges in navigation, switching between views, and understanding conversational context across branches. Overall, our results suggest that future LLM interfaces should move beyond linear chat toward interactive workspaces that support both exploration and synthesis, while balancing structure with usability.

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References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [2] Rifat Mehreen Amin, Oliver Hans Kühle, Daniel Buschek, and Andreas Butz. 2026. PromptCanvas: Composable Prompting Workspaces Using Dynamic Widgets for Exploration and Iteration in Creative Writing. *ACM Trans. Interact. Intell. Syst.* (June 2026). doi:10.1145/3817049 Just Accepted.
- [3] Ian Arawjo, Chelse Swoopes, Priyan Vaithilingam, Martin Wattenberg, and Elena L. Glassman. 2024. ChainForge: A Visual Toolkit for Prompt Engineering and LLM Hypothesis Testing. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 304, 18 pages. doi:10.1145/3613904.3642016
- [4] Shraddha Barke, Michael B. James, and Nadia Polikarpova. 2023. Grounded Copilot: How Programmers Interact with Code-Generating Models. *Proc. ACM Program. Lang.* 7, OOPSLA1, Article 78 (April 2023), 27 pages. doi:10.1145/3586030
- [5] John Brooke. 1996. SUS: A 'Quick' and 'Dirty' Usability Scale. In *Usability Evaluation in Industry*, Patrick W. Jordan, Bruce Thomas, Bernard A. Weerdmeester, and Ian Lyall McClelland (Eds.). Taylor and Francis, Chapter 21, 189–194.
- [6] Alex Bäuerle, Adam Connors, Alexander Novikov, Adam Zsolt Wagner, Ngân Vù, Fernanda Viegas, Martin Wattenberg, and Lucas Dixon. 2026. Intentionmaking and Sensemaking: Human Interaction with AI-Guided Mathematical Discovery. arXiv:2605.05921 [cs.AI] <https://arxiv.org/abs/2605.05921>
- [7] Daniela Fernandes, Steven Villa, Salla Nicholls, Otto Haavisto, Daniel Buschek, Albrecht Schmidt, Thomas Kosch, Chenxinran Shen, and Robin Welsch. 2026. AI makes you smarter but none the wiser: The disconnect between performance and metacognition. *Computers in Human Behavior* 175 (2026), 108779. doi:10.1016/j.chb.2025.108779
- [8] Tong Gao, Mira Dontcheva, Eytan Adar, Zhicheng Liu, and Karrie G. Karahalios. 2015. DataTone: Managing Ambiguity in Natural Language Interfaces for Data Visualization. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology* (Charlotte, NC, USA) (UIST '15). Association for Computing Machinery, New York, NY, USA, 489–500. doi:10.1145/2807442.2807478
- [9] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, 139–183.
- [10] Daphne Ippolito, Ann Yuan, Andy Coenen, and Sehmon Burnam. 2022. Creative Writing with an AI-Powered Writing Assistant: Perspectives from Professional Writers. doi:10.48550/ARXIV.2211.05030 Version Number: 1.
- [11] Peiling Jiang, Jude Rayan, Steven P. Dow, and Haijun Xia. 2023. Graphologue: Exploring Large Language Model Responses with Interactive Diagrams. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, San Francisco CA USA, 1–20. doi:10.1145/3586183.3606737
- [12] Evangelos Karapanos, John Zimmerman, Jodi Forlizzi, and Jean-Bernard Martens. 2009. User experience over time: an initial framework. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). Association for Computing Machinery, New York, NY, USA, 729–738. doi:10.1145/1518701.1518814
- [13] David Kirsh. 2010. Thinking with external representations. *AI & SOCIETY* 25, 4 (Nov. 2010), 441–454. doi:10.1007/s00146-010-0272-8
- [14] Mina Lee, Katy Ilonka Gero, John Joon Young Chung, Simon Buckingham Shum, Vipul Raheja, Hua Shen, Subhashini Venugopalan, Thiemo Wambsganss, David Zhou, Emad A. Alghamdi, Tal August, Avinash Bhat, Madiha Zahrah Choksi, Senjuti Dutta, Jin L.C. Guo, Md Naimul Hoque, Yewon Kim, Simon Knight, Seyed Parsa Neshaei, Antonette Shibani, Disha Shrivastava, Lila Shroff, Agnia Sergeyuk, Jessi Stark, Sarah Sterman, Sitong Wang, Antoine Bosselut, Daniel Buschek, Joseph Chee Chang, Sherol Chen, Max Kreminski, Joonsuk Park, Roy Pea, Eugenia Ha Rim Rho, Zejiang Shen, and Pao Siangliulue. 2024. A Design Space for Intelligent and Interactive Writing Assistants. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 1054, 35 pages. doi:10.1145/3613904.3642697
- [15] Jiaming Li, Peinuan Qin, and Yi-Chieh Lee. 2026. Branchat: A Tree-Structured Interface for Efficient Revisitation in Long-Horizon LLM Conversations. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems* (CHI EA '26). Association for Computing Machinery, New York, NY, USA, Article 137, 6 pages. doi:10.1145/3772363.3798792
- [16] Michael Xieyang Liu, Tongshuang Wu, Tianying Chen, Franklin Mingzhe Li, Aniket Kittur, and Brad A Myers. 2024. Selenite: Scaffolding Online Sensemaking with Comprehensive Overviews Elicited from Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 837, 26 pages. doi:10.1145/3613904.3642149
- [17] Tao Long, Sitong Wang, Emilie Fabre, Tony Wang, Anup Sathya, Jason Wu, Savvas Dimitrios Petridis, Ding Li, Tuhin Chakraborty, Yue Jiang, Jingyi Li, Tiffany Tseng, Ken Nakagaki, Qian Yang, Nikolas Martelaro, Jeffrey V Nickerson, and Lydia B Chilton. 2025. Facilitating Longitudinal Interaction Studies of AI Systems. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology* (UIST Adjunct '25). Association for Computing Machinery, New York, NY, USA, Article 13, 5 pages. doi:10.1145/3746058.3758469
- [18] Richard Mander, Gitta Salomon, and Yin Yin Wong. 1992. A “pile” metaphor for supporting casual organization of information. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Monterey, California, USA) (CHI '92). Association for Computing Machinery, New York, NY, USA, 627–634. doi:10.1145/142750.143055
- [19] Damien Masson, Sylvain Malacria, Géry Casiez, and Daniel Vogel. 2024. Direct-GPT: A Direct Manipulation Interface to Interact with Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 975, 16 pages. doi:10.1145/3613904.3642462
- [20] Srishti Palani, Zijian Ding, Austin Nguyen, Andrew Chuang, Stephen MacNeil, and Steven P. Dow. 2021. CoNote: Suggesting Queries Based on Notes Promotes Knowledge Discovery. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 726, 14 pages. doi:10.1145/3411764.3445618
- [21] Srishti Palani, Yingyi Zhou, Sheldon Zhu, and Steven P. Dow. 2022. InterWeave: Presenting Search Suggestions in Context Scaffolds Information Search and Synthesis. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 93, 16 pages. doi:10.1145/3526113.3545696
- [22] Peter Irolli and Stuart Card. 2005. The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis. In *Proceedings of international conference on intelligence analysis*, Vol. 5. McLean, VA, USA, 2–4.
- [23] Dimitri Popolov, Michael Callaghan, and Paul Luker. 2000. Conversation space: visualising multi-threaded conversation. In *Proceedings of the Working Conference on Advanced Visual Interfaces* (Palermo, Italy) (AVI '00). Association for Computing Machinery, New York, NY, USA, 246–249. doi:10.1145/345513.345330
- [24] Gonzalo Ramos, Napol Rachatasumrit, Jina Suh, Rachel Ng, and Christopher Meek. 2022. ForSense: Accelerating Online Research Through Sensemaking Integration and Machine Research Support. *ACM Trans. Interact. Intell. Syst.* 12, 4, Article 30 (Nov. 2022), 23 pages. doi:10.1145/3532853
- [25] Mohi Reza, Nathan M Laundry, Ilya Musabirov, Peter Dushniku, Zhi Yuan “Michael” Yu, Kashish Mittal, Tovi Grossman, Michael Liut, Anastasia Kuzminykh, and Joseph Jay Williams. 2024. ABScribe: Rapid Exploration & Organization of Multiple Writing Variations in Human-AI Co-Writing Tasks using Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 1042, 18 pages. doi:10.1145/3613904.3641899
- [26] George Robertson, Mary Czerwinski, Kevin Larson, Daniel C. Robbins, David Thiel, and Maarten van Dantzich. 1998. Data mountain: using spatial memory for document management. In *Proceedings of the 11th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, California, USA) (UIST '98). Association for Computing Machinery, New York, NY, USA, 153–162. doi:10.1145/288392.288596
- [27] Joey Scarr, Andy Cockburn, Carl Gutwin, and Sylvain Malacria. 2013. Testing the robustness and performance of spatially consistent interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 3139–3148.

- doi:10.1145/2470654.2466430
- [28] Ben Shneiderman. 1983. Direct manipulation: A step beyond programming languages. *Computer* 16, 08 (1983), 57–69.
 - [29] Ben Shneiderman. 1996. The eyes have it: A task by data type taxonomy for information visualizations. In *Proceedings 1996 IEEE symposium on visual languages*. IEEE, 336–343.
 - [30] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 644, 26 pages. doi:10.1145/3613904.3642400
 - [31] Sangho Suh, Bryan Min, Srishti Palani, and Haijun Xia. 2023. Sensecave: Enabling Multilevel Exploration and Sensemaking with Large Language Models. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 1, 18 pages. doi:10.1145/3586183.3606756
 - [32] John Sweller. 1988. Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science* 12, 2 (1988), 257–285. doi:10.1207/s15516709cog1202_4
 - [33] Lev Tankelevitch, Viktor Kewenig, Auste Simkute, Ava Elizabeth Scott, Advait Sarkar, Abigail Sellen, and Sean Rintel. 2024. The Metacognitive Demands and Opportunities of Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 680, 24 pages. doi:10.1145/3613904.3642902
 - [34] Michael Terry and Elizabeth D. Mynatt. 2002. Side views: persistent, on-demand previews for open-ended tasks. In *Proceedings of the 15th Annual ACM Symposium on User Interface Software and Technology* (Paris, France) (UIST '02). Association for Computing Machinery, New York, NY, USA, 71–80. doi:10.1145/571985.571996
 - [35] Tongshuang Wu, Ellen Jiang, Aaron Donsbach, Jeff Gray, Alejandra Molina, Michael Terry, and Carrie J Cai. 2022. PromptChainer: Chaining Large Language Model Prompts through Visual Programming. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 359, 10 pages. doi:10.1145/3491101.3519729
 - [36] Tongshuang Wu, Michael Terry, and Carrie Jun Cai. 2022. AI Chains: Transparent and Controllable Human-AI Interaction by Chaining Large Language Model Prompts. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 385, 22 pages. doi:10.1145/3491102.3517582
 - [37] Ryan Yen and Jian Zhao. 2024. Memolet: Reifying the Reuse of User-AI Conversational Memories. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) (UIST '24). Association for Computing Machinery, New York, NY, USA, Article 58, 22 pages. doi:10.1145/3654777.3676388
 - [38] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. 2022. Wordcraft: Story Writing With Large Language Models. In *Proceedings of the 27th International Conference on Intelligent User Interfaces* (Helsinki, Finland) (IUI '22). Association for Computing Machinery, New York, NY, USA, 841–852. doi:10.1145/3490099.3511105
 - [39] J.D. Zamfirescu-Pereira, Richmond Y. Wong, Bjoern Hartmann, and Qian Yang. 2023. Why Johnny Can't Prompt: How Non-AI Experts Try (and Fail) to Design LLM Prompts. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 437, 21 pages. doi:10.1145/3544548.3581388
 - [40] Zheng Zhang, Jie Gao, Ranjodh Singh Dhaliwal, and Toby Jia-Jun Li. 2023. VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 5, 30 pages. doi:10.1145/3586183.3606800

A Methodology

A.1 Participants

Table 2: Participant Overview (pre-task questionnaire Survey): Demographics, professional background, familiarity with LLM tools, usage frequency, and primary tasks.

Nr.	Gender	Prof. Writer	Job	Age	Familiarity	Frequency	LLM Tasks Used
P1	Female	No	PhD student	27	Very familiar	Daily	Writing/editing; coding
P2	Female	No	PhD Student	28	Very familiar	Multiple/day	Writing/editing; coding; data analysis; planning; brainstorming
P3	Male	No	Software Engineer	32	Moderately familiar	3–5/week	Writing/editing; coding; data analysis; learning
P4	Male	No	Accountant	29	Moderately familiar	3–5/week	Writing/editing; brainstorming; learning
P5	Male	Yes	Data Engineer	34	Expert	Multiple/day	Writing/editing; coding; data analysis; brainstorming; learning
P6	Female	No	Sales	29	Moderately familiar	3–5/week	Writing/editing; brainstorming; learning
P7	Female	Yes	Research Communication Specialist	29	Very familiar	1–2/week	Writing/editing; planning; brainstorming; learning
P8	Non-binary	No	Student	28	Very familiar	3–5/week	Writing/editing; coding; data analysis; learning
P9	Male	Yes	Researcher	33	Expert	Daily	Writing/editing; coding; brainstorming
P10	Male	No	Data Science Specialist	31	Very familiar	Multiple/day	Writing/editing; coding; data analysis; planning; brainstorming; creative writing
P11	Female	No	Master’s Student	26	Very familiar	Daily	Writing/editing; coding; brainstorming; learning
P12	Male	No	Visual Designer	27	Very familiar	Daily	Writing/editing; brainstorming
P13	Male	No	Unemployed	35	Very familiar	Daily	Coding; brainstorming; learning
P14	Male	No	Software Engineer	28	Very familiar	3–5/week	Coding; data analysis; planning; brainstorming; learning
P15	Male	No	CS Master Student	28	Very familiar	Multiple/day	Writing/editing; coding; data analysis; brainstorming; learning
P16	Male	No	Software Developer	29	Very familiar	Multiple/day	Coding; data analysis; brainstorming; learning
P17	Male	No	Intern (Full-Stack)	21	Very familiar	Daily	Writing/editing; coding; data analysis; planning; brainstorming; learning
P18	Male	No	Student	24	Very familiar	Daily	Writing/editing; coding; data analysis; planning; brainstorming; learning
P19	Male	No	Mechanical Eng. Intern	24	Very familiar	Multiple/day	Writing/editing; coding; data analysis; planning; brainstorming; learning
P20	Male	No	Student	25	Very familiar	Multiple/day	Writing/editing; coding; data analysis; planning; brainstorming; learning
P21	Male	No	CS Master Student	21	Expert	Multiple/day	Writing/editing; coding; data analysis; planning; brainstorming; learning; creative writing
P22	Male	No	Researcher	23	Very familiar	3–5/week	Writing/editing; coding
P23	Male	No	GenAI Researcher	54	Expert	Multiple/day	Writing/editing; data analysis; planning; brainstorming
P24	Male	No	Working Student (PM/Dev)	22	Very familiar	Multiple/day	Writing/editing; coding; data analysis; planning

B Results

B.1 NASA-TLX

B.2 Objective Telemetry

Table 3: The raw NASA-TLX results (N=24).

Factor	M	SD
Mental Demand	2.83	1.55
Physical Demand	1.50	1.10
Temporal Demand	1.96	1.16
Success	5.46	1.14
Effort	3.29	1.68
Annoy	1.67	0.82
Overall NASA-TLX (Sum)	16.71	4.44
Overall NASA-TLX (Mean)	2.79	0.74

Unpublished working draft.
Not for distribution.

Table 4: Objective telemetry data from the main study for CanvasConvo. Values represent per-participant averages.

Telemetry	Value
Messages	51.04
Conversations	6.88
Interaction distribution (%)	
Chat interactions	47.2%
Canvas interactions	28.3%
Sidebar navigation	11.4%
Session duration (minutes)	
Canvas sessions	34.10
Chat sessions	13.00
Branching behavior	
Branches created	3.29
Branch navigation events	3.88
Navigation events	
Timeline navigation	5.50
Canvas interaction events	
Zoom level changes	36.71
Node clicks	10.25
Canvas search usage	3.92
Timeline navigation clicks	5.50

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